

PLS-CADD Version 21.01x64 1:58:13 PM May 22, 2026
RR Power Consulting Inc. - Canada
Project Name: 'P:\Wilson\PLSCADD\PLS-CADD TRAINING\TEST SESSION 3.don'

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The use of IEEE Standard 738-2023 is experimental and not yet intended for production use.

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Steady-State Thermal Rating

IEEE Standard 738-2023 method of calculation

Weather Conditions

Air temperature:	40.000 (deg C)
Wind speed:	0.610 (m/s)
Wind to conductor angle:	90.000 (deg)
Atmosphere type:	Industrial

Solar Conditions

Date:	June 21 (day of the year with most solar heating)
Day of year:	172
Sun time:	11.000 (hours)
Altitude:	76.024 (deg)
Azimuth:	100.546 (deg)

Conductor Properties

Description:	155.4 kcmil 7/0 Strands ANAHEIM AAAC - Adapted from 1970's Publicly Available Data
Azimuth:	90.000 (deg)
AC resistance at 25.0 (deg C):	0.4330 (Ohm/km)
AC resistance at 75.0 (deg C):	0.5068 (Ohm/km)
Solar absorptivity:	0.500
Emissivity:	0.500
Outer diameter:	11.354 (mm)
Outer strand diameter:	0.000 (mm)
Outer strand layers:	0
Outer surface finish:	Smooth
Cable is thermal bimetallic:	False
Heat capacity:	205.849 (Watt-s/m-deg C)
Latitude:	26.800 (deg)

Elevation above sea level: 100.000 (m)

Analysis Results

Current:	559.979	(Amps)
Conductor surface temperature:	250.000	(deg C)
Conductor core temperature:	259.633	(deg C)
Convective cooling:	180.698	(Watt/m)
Radiative cooling:	65.997	(Watt/m)
Solar heating:	4.600	(Watt/m)
Equivalent global solar radiation:	810.328	(Watt/m^2)
Final conductor azimuth:	90.000	(deg)
Final solar absorptivity:	0.500	
Final emissivity:	0.500	
Final wind to conductor angle:	90.000	(deg)

